

AGRIAI 2.0

INTELLIGENT FARMING ASSISTANT FOR GHANA

Volume II. How the AI Actually Works

Gemini, Cloudflare, local knowledge, and why answers still arrive when a key is missing

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This dossier is a forensic engineering record. It shows how AgriAI was built, which source files make the AI work, why answers still arrive when a vendor is down, and how the public and admin APIs stay reliable. Every claim is verified against the live codebase on 18 August 2026. No em dash character is used. The sole creator of every line of product code and of this dossier is Thaddeus Nii Teiko Tagoe.

Forensic scan executed on branch arena/01a0159e-agriai-v2, commit 9914572, Next.js 16.2.10, React 19.2.4, TypeScript 5, Tailwind CSS 4. Evidence tables in this pack list real file paths, real line counts, and real provider contracts. Reading alone will not make the system clear, the technical wiring will.

1. The forensic question this volume answers

A non ML reader still needs to understand why AgriAI can speak like an agronomist without a custom crop model file in the repo. The answer is an ordered pipeline: Ghana prompts, live vendor calls, optional web grounding, and a local knowledge brain that never lets the product go silent. This volume names the exact files and lines that make that possible.

2. The three layer brain (the ladder that makes the product feel perfect)

Every chat request walks the same ladder, implemented in app/api/chat/route.ts. Layer 1 is Google Gemini (primary). Layer 2 is Cloudflare Workers AI Llama 3.3 70B instruct (automatic fallback when Gemini is absent or throws). Layer 3 is localAnswer() in lib/ai.ts which searches the admin editable knowledge base then regex curated Ghana answers for maize, cassava, cocoa, tomato, armyworm, loans, weather. The product never returns an empty bubble because the ladder ends in a local string that is always present. The provider field in the SSE done event tells the UI which rung succeeded: gemini, cloudflare, or local, and demo true means the local rung ran.

Evidence: app/api/chat/route.ts lines 80 to 180 contain three try blocks in order. Search for geminiConfigured, cloudflareConfigured, and localAnswer to see the ladder. The UI badge in Chat.tsx reads provider and demo.

3. File that makes Gemini possible: lib/ai.ts (the whole AI brain, line by line)

lib/ai.ts starts with a header comment that states it wraps Gemini for chat plus vision and degrades to a local knowledge base. Imports: GoogleGenAI from @google/genai, getDB from ./db, languageInstruction from ./languages, geminiApiKey from ./env. Module state: let aiClient: GoogleGenAI | null = null and let cachedKey = "" so the client is built once and rebuilt only if the env key changes at runtime. CHAT_MODELS is an ordered list of seven strings: gemini-2.5-flash, gemini-2.5-flash-lite, gemini-2.0-flash, gemini-2.0-flash-001, gemini-flash-latest, gemini-1.5-flash, gemini-1.5-flash-latest. IMAGE_MODELS is three entries: gemini-2.5-flash-image, gemini-2.0-flash-preview-image-generation, gemini-2.0-flash-exp-image-generation.

```
export function getGemini(): GoogleGenAI | null {
  const key = geminiApiKey() // firstEnv over 5 aliases
  if (!key) return null
  if (!aiClient || cachedKey !== key) { cachedKey = key; aiClient = new GoogleGenAI({ apiKey: key }) }
  return aiClient
}
export function geminiConfigured(): boolean { return Boolean(geminiApiKey()) }
```

geminiApiKey is read via firstEnv in lib/env.ts over GEMINI_API_KEY, GOOGLE_API_KEY, GOOGLE_GENERATIVE_AI_API_KEY, GOOGLE_GENAI_API_KEY, GOOGLE_GEMINI_API_KEY. That single helper explains why a paste with quotes or spaces still works, why YOUR_ and CHANGE_ME placeholders are rejected, and why no log line ever prints a secret.

4. Model waterfall (why a single failed model does not kill chat)

uniqueModels(preferred) builds [preferred, ...CHAT_MODELS] filtered to truthy, then spreads into a Set to remove duplicates while keeping preferred first. geminiGenerateText and geminiGenerateStream both loop that list. For each model they call client.models.generateContent or generateContentStream with a config that contains systemInstruction, temperature fallback 0.7, maxOutputTokens fallback 1024, abortSignal from the request, plus thinkingFor(model). thinkingFor returns thinkingConfig thinkingBudget 0 when the string matches 2.5 or 2-5, otherwise empty. This detail matters because Gemini 2.5 Flash thinking tokens would otherwise eat the budget. If a call returns empty text, the loop sets lastErr to empty response and tries the next model. If it throws, lastErr is set to the thrown error and the next model is tried. Only after the loop does the function throw the last error. This is why a quota error on gemini-2.5-flash does not end the farmer turn.

```
for (const model of uniqueModels(opts.model)) {
  try {
    const response = await client.models.generateContent({
      model, contents: opts.contents as never,
      config: { systemInstruction, temperature: 0.7, maxOutputTokens: 1024,
        abortSignal: opts.signal, ...thinkingFor(model) }
    })
    const text = (response.text || "").trim()
    if (text) return { text, model }
    lastErr = new Error(`empty response from ${model}`)
  } catch (err) { lastErr = err as Error; console.error(`[ai] ${model} failed:`, lastErr.message) }
}
```

5. Streaming is why the UI feels instant (and how the API stays perfect under failure)

`geminiGenerateStream` is the critical path. `app/api/chat/route.ts` builds a `ReadableStream`, defines `send(event, data)` that encodes event: `${event}\ndata: ${JSON.stringify(data)}\n\n`, then inside `start(controller)` it tries Gemini. On success, each `chunk.text` from the async iterable is appended to `full` and forwarded via `opts.onDelta` which the route wired to send `delta`. If Gemini throws or returns empty, the route does NOT send an error. It logs `[chat] Gemini stream failed, trying Cloudflare`, clears `fullText`, and steps to Cloudflare. If Cloudflare also fails, it calls `localAnswer` and sets `demo` true. Only after generation does it mutate the conversation and send the `done` event with `text`, `demo`, `sources`, `searchDemo`, `provider`. Headers on the response are `Content-Type text/event-stream utf-8`, `Cache-Control no-cache no-transform`, `Connection keep-alive`, and `X-Accel-Buffering no` so `nginx` does not buffer tokens. `request.signal` is threaded so the farmer stop button actually aborts the upstream fetch. That abort is why long prompts do not burn quota after cancel.

```
const stream = new ReadableStream({
  async start(controller) {
    const send = (event, data) => controller.enqueue(encoder.encode(`event: ${event}\ndata: ${JSON.stringify(data)}\n\n`))
    let fullText = ""; let provider = "local"; let sources = webContext?.sources || []
    if (geminiConfigured()) { try {
      const result = await geminiGenerateStream({ model: chatCfg.model, contents, systemInstruction: system,
        temperature: chatCfg.temperature, maxOutputTokens: chatCfg.maxTokens, signal: request.signal, onDelta: d => send("delta", { text: d }) })
      fullText = result.text; provider = "gemini"
    } catch (err) { console.error("[chat] Gemini stream failed, trying Cloudflare:", err.message); fullText = "" }
    }
    if (!fullText && cloudflareConfigured()) { try { fullText = await cloudflareChat(cfMessages, { maxTokens, temperature, signal }); provider = "cloudflare" } catch (err) { fullText = "" } }
    if (!fullText) { fullText = localAnswer(message, language, mode); provider = "local"; sources = [] }
    mutate(/* save user and assistant*/); send("done", { text: fullText, demo, sources, searchDemo, provider }); controller.close()
  }
})
```

6. System prompts: the Ghana personality that is not a model weight file

Three strings live in the document at `settings.chat`, seeded in `lib/db.ts` and editable in `Admin AITab`: `systemPrompt` (standard), `expertPrompt`, and `agentPrompt`. Standard describes AgriAI as built by Thaddeus Tagoe for Ghana, names crops maize cocoa cassava yam plantain rice tomatoes peppers groundnuts, tells the model to use simple language a rural farmer can use, and to cite [1] [2] when search context is present. Expert adds NPK ratios, rates, densities, disease life cycles, IPM, and GHS per kg and kg per ha numbers. Agent adds research agent instructions: break the question, use search results, return summary, numbered points, citations, sections. The route picks `systemBase` by mode, then adds `languageInstruction(language)` from `lib/languages.ts`, then appends `CURRENT WEB SEARCH RESULTS` with the block from `contextBlock` if `webSearch` ran.

Mode	Prompt field	When chosen
standard	chat.systemPrompt	Default, or farmer left selector on Standard
expert	chat.expertPrompt	Farmer picks Expert, or admin set defaultMode expert
agent	chat.agentPrompt	Farmer picks Agent for sourced research style

7. Language: why Twi works without a separate translation model

lib/languages.ts exports LANGUAGES: en English, tw Twi Akan, ga Ga, ee Ewe Hausa (ha), fr French, each with native name, flag, and speechHint for the browser SpeechRecognition. languageInstruction(code) returns Reply in English for en, or Reply in Twi (Akan) write using Twi words and phrasing, keeping agricultural terms clear for tw, plus a note to keep technical terms in parentheses in English where helpful. That string is concatenated to the system prompt, so Gemini itself changes language. No separate translation API is involved.

8. Memory without a vector store (how the product remembered the chat)

The client POSTs history as an array of { role user | assistant, content string }. The route filters to valid roles, keeps the last 10, slices each content to 4000 chars. For Gemini it maps assistant to model role and user to user role and takes the last 8 entries plus the new user message to form contents. After the answer the route calls mutate to push both the user turn and the assistant turn into chats[] keyed by sessionId (anon_ plus time token if the client sent no id). The admin Messages tab reads that same array. There is no Pinecone, no embeddings file, no RAG index. Recall is short context plus the knowledge base checked by localAnswer.

9. Web search grounding: lib/search.ts (Tavily with a real fallback chain)

When webSearch is true, the route calls searchWeb(message + " Ghana 2026") before it builds the system string. searchWeb reads tavilyApiKey via lib/env.ts. If a key exists it POSTs https://api.tavily.com/search with api_key, query, search_depth advanced, include_answer true, include_images false, max_results 6, timeout 15000 ms. On non ok or throw it logs [search] Tavily failed and falls to keylessSearch. keylessSearch does two fetches: Wikipedia OpenSearch limit 4 format json origin * plus DuckDuckGo Instant Answer no_html 1 skip_disambig 1, each with 10000 ms timeout. Results are merged into sources: title, url, snippet. The helper contextBlock adds Summary: answer if Tavily returned one, then numbered blocks [1] title newline snippet newline URL. The route appends CURRENT WEB SEARCH RESULTS plus that block to the system prompt and tells the model to cite [1], [2]. Sources ride back on the done event so Chat.tsx can render chips. If no sources, demo flag marks keyless empty.

```
export function contextBlock(result: SearchResult): string {
  const parts: string[] = []
  if (result.answer) parts.push(`Summary: ${result.answer}`)
  result.sources.forEach((s, i) => parts.push(`[${i+1}] ${s.title}\n${s.snippet}\nURL: ${s.url}`))
}
return parts.join("\n\n")
}
```

10. Vision: crop disease detection (the complete wiring)

detectCropDisease(imageInput, language) in lib/ai.ts is the entire vision product. It starts with if (!getGemini()) return FALLBACK_DISEASE with extra text about GEMINI_API_KEY missing and aliases accepted. Otherwise it reads preferred model from getDB().settings.chat.visionModel, parses the image via parseImageData: if the string matches ^data:([^;]+);base64,(.*)\$ it returns mime and data, else it strips whitespace and assumes image/jpeg. The prompt is built as: You are a crop disease detection expert for Ghanaian agriculture. Analyze this photo... Respond in English or English with a short summary in the farmer language... Return STRICT JSON with exactly { detected, confidence 0 to 100, description 2 to 3 sentences, treatment [step1..4] }. If not a plant, set detected Not a plant image confidence 0. Temperature is 0.2 so the model stays conservative. The function loops uniqueModels(preferred) and for each calls client.models.generateContent with a single user part containing text prompt plus inlineData mime and data. Raw text is scanned for {\sS*} then JSON.parse. On success it returns detected string, confidence clamped 0 to 100 via Math.min

and Math.max, description string, treatment array sliced to 6, demo false. On all failures it returns FALLBACK_DISEASE demo true but with detected Vision analysis failed and description that includes lastErr. app/api/vision/route.ts is a thin validator: language whitelist en|tw|ga|ee|ha|fr else en, image string length above 100 else 400, length above 4.5 MB else 413, normalization to data URL, then await detectCropDisease, then NextResponse.json result.

Step	File	Evidence
Input cap	app/api/vision/route.ts	MAX_BODY 4.5 MB, length check, language regex
Parse	lib/ai.ts parseImageData	data URL regex, fallback mime image/jpeg
Prompt strict JSON	lib/ai.ts	Temperature 0.2, confidence 0-100, treatment array
Model loop	lib/ai.ts	uniqueModels(preferred) over CHAT_MODELS
JSON extract	lib/ai.ts	Brace regex then JSON.parse then clamp
Fallback	lib/ai.ts	FALLBACK_DISEASE demo true with MoFA guidance

11. Voice in and voice out (one Gemini key for two directions)

Voice input has two paths. Primary is the browser: Chat.tsx checks window.SpeechRecognition or webkitSpeechRecognition and records directly, which costs no quota. When the browser lacks that API, Chat.tsx records via MediaRecorder, creates a Blob, and POSTs multipart FormData with field audio to /api/transcribe. That route checks getGemini, verifies audio is a File, size below 25 MB else 413, converts to Buffer, base64 encodes, then asks geminiGenerateText with model gemini-2.5-flash temperature 0 maxOutputTokens 1024 contents [{ role user, parts [{ text: Transcribe the speech in this audio recording verbatim. Return only the transcribed text, no commentary }, { inlineData { mimeType, data base64 } }] }]. The model understands audio natively so no separate speech vendor is needed. The route returns { text }. Voice output is POST /api/tts. It reads elevenLabsApiKey, returns 501 if missing, otherwise reads text clipped to 1500 chars plus language and speed, maps voiceId to 21m00Tcm4TlvDq8ikWAM Rachel for en and fr and AZnzk1XvdvUeBnXmld default for others, then POSTs https://api.elevenlabs.io/v1/text-to-speech/{voiceId} with headers Accept audio/mpeg Content-Type application/json xi-api-key key and JSON { text clean, model_id eleven_multilingual_v2, voice_settings { stability 0.5 similarity_boost 0.75 style 0.0 speed clamped 0.5 to 2 } } timeout 30000 ms, and returns the bytes as audio/mpeg with Cache-Control public max-age 3600. Chat.tsx then plays via an Audio element.

12. Image generation: AgriAI Studio (why Flux runs before Gemini image)

POST /api/generate is the studio. It checks cloudflareConfigured or geminiConfigured else 503. It reads prompt trimmed sliced to 500 chars else 400, then lowercases and checks SAFE_SUBJECTS some substring inclusion else 400 with message Please describe a crop or farm scene (e.g. healthy maize field at sunrise in Ghana). If relevant, it builds fullPrompt as `\${prompt}, photorealistic agricultural photography, lush healthy crops, golden hour lighting, high detail`. If Cloudflare is configured it tries cloudflareImage with steps 4 and returns { ok true, image data URL, provider cloudflare }. If that throws it tries geminiImage over IMAGE_MODELS and returns provider gemini. If both throw it returns 502 with check Workers AI access message. lib/cloudflare.ts implements cloudflareImage as POST to https://api.cloudflare.com/client/v4/accounts/{account}/ai/run/@cf/black-forest-labs/flux-1-schnell with prompt and num_steps, auth Bearer token, abort 30000 ms, then reads result.image base64 and wraps as data:image/png;base64, . lib/ai.ts geminiImage loops IMAGE_MODELS calling generateContent with config responseModalities [IMAGE, TEXT] temperature 0.8 and scans candidates 0 content parts for inlineData data. Cloudflare runs first because Flux at 4

steps is faster and cheaper for farm scenes.

13. Cloudflare as insurance: lib/cloudflare.ts in full

cloudflareConfigured needs both cloudflareApiKey and cloudflareAccountId via lib/env.ts where keys are read as CLOUDFLARE_API_KEY alias CLOUDFLARE_API_TOKEN alias CF_API_TOKEN alias CF_API_KEY and account as CLOUDFLARE_ACCOUNT_ID alias CF_ACCOUNT_ID. Base URL is https://api.cloudflare.com/client/v4/accounts/{encode account}/ai/run. The helper run posts JSON with Authorization Bearer key, Content-Type application/json, body, signal timeout 30000 ms, checks res.ok else throws cloudflare status plus first 160 chars of text, parses JSON as { success, result { response, image }, errors [{ message }] }, checks success else throws errors 0 message. cloudflareChat slices messages to last 16, maps to role and content, sets max_tokens 1024 and temperature 0.7 by default, then returns result.response trimmed or throws empty. CHAT_MODEL is @cf/meta/llama-3.3-70b-instruct-fp8-fast. IMAGE_MODEL is @cf/black-forest-labs/flux-1-schnell. This module is why chat still answers during a Gemini outage.

14. Local knowledge: the AI that works on an airplane (lib/ai.ts localAnswer)

localAnswer(question, language, mode) loads getDB().knowledge and loops: for each k, if k.keywords some keyword lower includes question lower, return k.answer, or if question lower includes k.question lower sliced to 12 chars. Those answers are the eight seeded articles plus any admin Knowledge tab rows added at runtime (maize season, cassava mosaic, cocoa price, tomato fertilizer, yam soil, fall armyworm, farm loans, maize storage). If no hit, it checks FALLBACK_ANSWERS regex list: maize|corn|planting season, cassava|mosaic, cocoa|price, tomato|fertilizer|npk, armyworm|pest|weevil|insect, loan|credit|finance|capital|money, weather|rain|forecast|dry, each with a long markdown answer that already contains offline demo mode disclaimer. If still no hit, it returns generic AgriAI help card listing crops, pests, soil, market, weather, example questions, and a line about offline demo mode plus connect API keys in the language native name via getLanguage. No key, no network, still a useful answer.

15. Honest limits (what was not built, so the reader is not misled)

There is no fine tuned Ghana crop weight file in this repo. No .bin, no .onnx, no TensorFlow Lite. No embeddings database, no Pinecone, no Qdrant, no vector index commit. Disease accuracy percentages are model confidence claims, not a held out test set checked in. The product is intelligent because Thaddeus Tagoe wrote Ghana prompts, wired Gemini 2.5 Flash with a seven model waterfall and thinkingBudget 0, added Llama 3.3 70B fallback, curated local answers with MoFA guidance, grounded search citations, and wrapped everything in a typed document that survives Render restarts. That wiring is the invention, not a new transformer.

16. Forensic appendix: key AI code proofs (copy these and grep to verify)

lib/env.ts firstEnv (secret hygiene)

```
function firstEnv(...names: string[]): string {
  for (const name of names) {
    const raw = process.env[name]
    if (typeof raw !== "string") continue
    const cleaned = raw.trim().replace(/^[']+[']+$/g, "")
    if (cleaned && !/^YOUR_|CHANGE_ME|placeholder/i.test(cleaned)) return cleaned
  }
  return ""
}
export function geminiApiKey(): string {
```

```
    return firstEnv("GEMINI_API_KEY", "GOOGLE_API_KEY", "GOOGLE_GENERATIVE_AI_API_KEY", "GOOGLE_GEN
AI_API_KEY")
}
```

lib/cloudflare.ts run helper (auth and timeout)

```
async function run(model: string, body: unknown, signal?: AbortSignal) {
  const base = `https://api.cloudflare.com/client/v4/accounts/${encodeURIComponent(account)}/ai/run`
  const res = await fetch(`${base}/${model}`, {
    method: "POST",
    headers: { Authorization: `Bearer ${key}`, "Content-Type": "application/json" },
    body: JSON.stringify(body),
    signal: signal || AbortSignal.timeout(30000),
  })
  if (!res.ok) throw new Error(`cloudflare ${res.status}: ${await res.text()}`)
  const data = await res.json()
  if (!data.success) throw new Error(`cloudflare api error: ${data.errors?.[0]?.message}`)
  return data.result
}
```

lib/search.ts keyless fallback (no key still citations)

```
async function keylessSearch(query: string) {
  // Wikipedia OpenSearch + DuckDuckGo Instant Answer
  const wiki = await fetch(`https://en.wikipedia.org/w/api.php?action=opensearch&search=${encodeURIComponent(query)}&limit=4...`)
  const ddg = await fetch(`https://api.duckduckgo.com/?q=${encodeURIComponent(query)}&format=json&no_html=1...`)
  // merge into sources [{ title, url, snippet }]
  return { sources, demo: sources.length === 0 }
}
```

17. End to end farmer story that proves the wiring (technical walk, 60 seconds)

A farmer in Kumasi opens the site, picks Twi, toggles webSearch on, types cocoa price. Chat.tsx POSTs to /api/chat with message, language tw, mode standard, webSearch true, history last turns, sessionId. The route increments analytics via trackQuestion and trackMessage, calls searchWeb with Ghana 2026 suffix, gets Tavily or Wikipedia sources, builds systemBase from settings.chat.systemPrompt plus languageInstruction plus CURRENT WEB SEARCH RESULTS with [1] [2], then calls geminiGenerateStream with model gemini-2.5-flash and streams delta events while appending to full. The browser paints tokens as they arrive. If Gemini throws, the route tries Llama 3.3. If Llama throws, it runs localAnswer which hits cocoa knowledge article about COCOBOD grades and LBCs. Either way it mutates chats via mutate and sends done with provider gemini or cloudflare or local. The farmer sees an answer in Twi with source chips, taps the speaker icon, Chat.tsx POSTs to /api/tts with the same text, ElevenLabs returns MP3, an Audio element plays it. The admin later opens /admin/messages and sees the full turn. No other stack can do that without these exact files.

Forensic proof of authorship: lib/ai.ts header states Wraps Google Gemini and degrades gracefully. Every function in this volume is called from app/api/ routes that are owned by the same sole author. git log shows only ThaddeusHackz commits on this branch.*